

Lais Borbolato

📍 Universidade de São Paulo - Instituto de Astronomia, Geofísica e Ciências Atmosféricas (IAG)

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Education

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|------------|---|-------------|
| PhD | University of São Paulo , Ph.D. in Astronomy, Universidade de São Paulo | 2023 - 2027 |
| | <ul style="list-style-type: none"> • Advisor: Prof. Dr. Silvia Rossi, co-advisor: Prof. Dr. Hélio D. Perottoni • Thesis title: <i>The Milky Way's disk structures as a memory of its turbulent past</i> | |
| BSc | University of São Paulo , BSc in Astronomy, Universidade de São Paulo | 2019 - 2022 |
| | <ul style="list-style-type: none"> • Advisor: Prof. Dr. Silvia Rossi, co-advisor: Prof. Dr. Hélio D. Perottoni • Thesis title: <i>Kinematics and Detailed Chemistry for Monoceros Overdensity.</i> | |

Work experience

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| Ph.D. | | 2024 - 2027 |
| | <ul style="list-style-type: none"> • Funding by São Paulo Research Foundation (FAPESP) - Total: 198,000 BRL • Project: The Milky Way's disk structures as a memory of its turbulent past | |
| Ph.D. | | 2023 - 2024 |
| | <ul style="list-style-type: none"> • Funding by Brazilian Federal Agency for Support and Evaluation of Graduate Education (CAPES) - Total: 79,200 BRL • Project: The Milky Way's disk structures as a memory of its turbulent past | |
| Undergraduate Research | | 2021 - 2022 |
| | <ul style="list-style-type: none"> • Funding by São Paulo Research Foundation (FAPESP) - Total: 7,200 BRL • Project: Kinematics and Detailed Chemistry for Monoceros Overdensity | |
| Undergraduate Research | | 2020 - 2021 |
| | <ul style="list-style-type: none"> • Funding by Institutional Scientific Initiation Scholarship Program of the National Council for Scientific and Technological Development (PIBIC/CNPq) - Total: 4,800 BRL • Project: Chemodynamical properties of the Monoceros Stellar Overdensity | |
| Unified scholarship program (PUB) | | 2019 - 2020 |
| | <ul style="list-style-type: none"> • Total: 4,800 BRL • Customer service at the Abrahão de Moraes Observatory in Valinhos, São Paulo | |

Teaching and Outreach

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|--------------------------|---|------|
| Hands-on session: | How to communicate your research. | 2025 |
| | <ul style="list-style-type: none"> • This course was offered at IAG/USP to undergraduate Astronomy students, with the aim of teaching them how to disseminate their scientific work. | |
| Teaching: | Preparatory classes for the Brazilian Astronomy Olympiad. | 2026 |
| | <ul style="list-style-type: none"> • Classes were taught at the Etapa school for high school students. | |
| Teaching: | Preparatory classes for the Brazilian Astronomy Olympiad | 2025 |
| | <ul style="list-style-type: none"> • Classes were taught at the Liceu Pasteur Start Anglo school for elementary and high school students, totaling 25 hours. | |

Hands-on session: Astronomical data visualization with Matplotlib for Brazilian students	2025
This course was offered at IAG/USP for undergraduate Astronomy students, aiming to help them create clearer and more visually engaging figures for their academic productions.	
Teaching assistant: Astronomy for Teachers (IAG-USP)	2024
This advanced training course is designed to offer Brazilian high school physics teachers an introductory overview of various fields of astronomy, presented from a modern and up-to-date perspective.	
Astronomical Service at IAG-USP	2022 - 2024
Supervisor of guided visits at IAG-USP and teaching assistant in virtual sessions with public school students.	
Night with the Stars - Abrahão de Moraes Observatory	2019 - 2023
Monitor assisting with customer service at the Abrahão de Moraes Observatory (OAM) in Valinhos, São Paulo, including assembling telescopes and leading hands-on workshops with children.	
ASTROMINAS (website)	2020 - 2021
This project is dedicated to promoting Science and Technology among female high school students, helping to facilitate their access to university.	
Cecília project (website)	2020 - 2021
This project is dedicated to disseminating and popularizing Earth and Universe sciences in schools.	

Learning and Service

Organizer: "Astronomy at Noon" Seminar Series	2024 - 2025
Cycle of Seminars of the Astronomy Department of IAG/USP every Thursday at noon. It is dedicated to undergraduate students in exact sciences	
Organizer: Milky Way Brazilian Group (MWBR)	2024 - 2025
Member: Brazilian Astronomical Society (SAB)	2019 - 2025
Students representative: Astronomy Program Coordinating Committee (IAG-USP)	2024
Students representative: Defense of Diversity, Human Rights and Democracy Committee, CDDDHD (IAG-USP)	2021 - 2022
Students representative: IAG-USP Council	2020 - 2021
Organizer: Brazilian Astronomy Olympiad (OBA)	2019 - 2020

Publications

1st Author Publications - Journal Articles

<i>The Low-α Splash Population in the Milky Way</i>	2026
Borbolato, L.; Amarante, J A. S.; Perottoni, H. D.; Rossi, S.; Debattista, V. P.; Zhao-Yu, L.; Deg, N.; Khachataryants, T.; 2026, submitted. arXiv	
<i>Early Co-formation of the Milky Way's Thin and Thick Disks at Redshift $z > 2$</i>	2025
Borbolato, L.; Rossi, S.; Perottoni, H. D.; Limberg, G.; Amarante, J A. S.; Queiroz, A. B. A.; Chiappini, C.; Anders, F.; Santucci, R. M.; Barbosa, F. O.; Nogueira-Santos, J. V.; 2025, ApJ, 994, 126. DOI: 10.3847/1538-4357/ae0c96	

All-sky Kinematics and Chemistry of Monoceros Stellar Overdensity 2024
Borbolato, L.; Perottoni, H. D.; Rossi, S.; Limberg, G.; Pérez-Villegas, A.; Anders, F.; Antoja, T.; Laporte, C. F. P.; Rocha-Pinto, H. J.; Santucci, R. M.; 2024, ApJ, 960, 52.
 DOI: [10.3847/1538-4357/ad02fb](https://doi.org/10.3847/1538-4357/ad02fb)

Co-authored Publications - Journal Articles

Characterizing the velocity anisotropy of the Milky Way's stellar halo 2026
 Barbosa, F. O.; Pérez-Villegas, A.; Rossi, S.; Santucci, R. M.; Aguilar, L.; Perottoni, H. D.; Limberg, G.; **Borbolato, L.**; Nogueira-Santos, J. V. 2026, submitted. [arXiv](#)

GASTRO library II: Exploring Chemical Bimodalities in Disk Galaxies with GSE-like Mergers and Massive Star-forming Clumps 2025
 Amarante, J. A. S.; Laporte, C. F. P.; Debattista, V. P.; Beraldo e Silva, L.; Limberg, G.; Perottoni, H. D.; Li, Z.; **Borbolato, L.**; Fiteni, K.; Cao, C.; Deg, N.; Khachaturyants, T.; Liao, X.; 2025, submitted. [arXiv](#)

Extending the Chemical Reach of the H3 Survey: Detailed Abundances of the Dwarf-galaxy Stellar Stream Wukong/LMS-1 2024
 Limberg, G.; Ji, A. P.; Naidu, R. P.; Chiti, A.; Rossi, S.; Usman, S. A.; Ting, Y.; Zaritsky, D.; Bonaca, A.; **Borbolato, L.**; Speagle, J. S.; Chandra, V.; Conroy, C.; 2024, MNRAS, 530, 3.
[10.1093/mnras/stae969](https://doi.org/10.1093/mnras/stae969)

Searching for Extragalactic Exoplanetary Systems: The Curious Case of BD+20 2457 2021
 Perottoni, H. D.; Amarante, J. A. S.; Limberg, G.; Rocha-Pinto, H. J.; Rossi, S.; Anders, F.; **Borbolato, L.**; 2021, ApJL, 913, L3..
[10.3847/2041-8213/abfb06](https://doi.org/10.3847/2041-8213/abfb06)

Conferences, Workshops & Contributed talks

Talk	XLVIII Annual Meeting of the Brazilian Astronomical Society (Caxambu, Brazil)	Oct 2025
	• Title: <i>Thin disk splash population: evidence for early chemical thin disk formation in the Milky Way</i>	
Talk	IAG Science Day (São Paulo, Brazil)	Sep 2025
	• Title: <i>The formation of the ancient Milky Way disk</i>	
Talk	Deciphering the Cosmic Code for Galaxy Formation (Puerto Varas, Chile)	Dec 2024
	• Title: <i>Deciphering the old Milky Way thin disk formation</i>	
Poster	IAU Symposium 395: Stellar populations in the Milky Way and beyond (Paraty, Brazil)	Nov 2024
	• Title: <i>The Early and Fast Settlement of Milky Way Disk</i>	
Talk	XLVII Annual Meeting of the Brazilian Astronomical Society (Águas de Lindóia, Brazil)	Sep 2024
	• Title: <i>Deciphering the earliest stages of the Milky Way disk formation</i>	
Talk	XXXII IAU General Assembly (Cape Town, South Africa)	Aug 2024
	• Title: <i>The Early and Fast Settlement of Milky Way Disk</i>	
Talk	XVII Latin American Regional IAU Meeting (Montevideú, Uruguai)	Nov 2023
	• Title: <i>All-sky Kinematics and Chemistry of Monoceros Stellar Overdensity</i>	
Talk	XLVI Annual Meeting of the Brazilian Astronomical Society (Rio de Janeiro, Brazil)	Oct 2023
	• Title: <i>Kinematics and Detailed Chemistry for Monoceros Overdensity</i>	

Poster 75th Annual Meeting of the Brazilian Society for the Advancement of Science (Curitiba, Brazil)	Jun 2023
• Title: <i>Kinematics and Detailed Chemistry for Monoceros Overdensity</i>	
Poster 30th International Symposium on Scientific and Technological Initiation at USP (São Paulo, Brasil)	Oct 2022
• Title: <i>Kinematics and Detailed Chemistry for Monoceros Overdensity</i>	
Poster XLV Annual Meeting of the Brazilian Astronomical Society (Virtual)	Sep 2022
• Title: <i>Kinematics and Detailed Chemistry for Monoceros Overdensity</i>	
Poster 29th International Symposium on Scientific and Technological Initiation at USP (Virtual)	Oct 2021
• Title: <i>Chemodynamical properties of the Monoceros Stellar Overdensity</i>	
Poster XLIV Annual Meeting of the Brazilian Astronomical Society (Virtual)	Sep 2021
• Title: <i>Chemodynamical properties of the Monoceros Stellar Overdensity</i>	

Honors and Awards

Dr. Abrahão Aun Medal for outstanding scientific activities in the municipality of Vinhedo, SP, Brazil	2025
Honorable Mention for PhD Research Presentation at the X IAG Science Day	2025
• Title: The formation of the ancient Milky Way disk	
Work awarded at the 75th Annual Meeting of the Brazilian Society for the Advancement of Science (SBPC)	2023
• Title: Chemodynamical properties of the Monoceros Stellar Overdensity	
Honorable mention at the 30th International Symposium on Scientific and Technological Initiation at USP (SIICUSP)	2022
• Title: Chemodynamical properties of the Monoceros Stellar Overdensity	
Gold medal at the Brazilian Astronomy and Astronautics Olympiad (OBA)	2018
Gold medal at the Brazilian Astronomy and Astronautics Olympiad (OBA)	2017
Honorable mention at the Brazilian Public School Mathematics Olympiad (OBMEP)	2015

Skills

Languages: Portuguese (native), English

Programming: Python (main), R-project

Other resources: TOPCAT (large data sets), AGAMA/galpy/Gala (dynamics)